

Assignment #4

Building your first function

Your task

Write a function to compute a biodiversity index - we will use Simpson's Index (D) as an example

"Simpson's Index (D) measures the probability that two individuals randomly selected from a sample will belong to the same species"

Value is between 0 and 1, with lower values associated with *lower* diversity

See [Simpson Biodiversity](#)

- there are two versions you can pick either
 - $D = \sum(n_i(n_i - 1)) / (N(N - 1))$
 - * where n_i is the number of individuals of each species
 - * N is the total number of individuals of all species
 - $D = \sum((n_i/N)^2)$
 - * where n_i is the number of individuals of each species
 - * N is the total number of individuals of all species

Steps

1. Define the contract
 - in a quarto document describe inputs, outputs and what the function will do
2. Implement the function in R
 - power_gen_orig is a good template to start with - copy and modify
 - make sure you include documentation at the top for now follow style from power_gen_orig but you can be more descriptive if you want - as long as line starts with #, it is a comment and will not be executed as code

- save this file as a name.R file
 - *hint* use factors to make it easier to count the number of individuals of each species
3. Test the function with some example data that you generate
- include you test code in the quarto document

Do this one on your own

Put both the function and quarto in directory on your github site and submit the link on Canvas

Due Thursday May 14 8am

Grading Rubric

Coding R function (30 points)

- 10 points: function is in correct format
- 10 points: function runs without error
- 10 points: function has informative documentation

Quarto document (20 points)

- 10 points: descriptions of inputs, outputs and what the function does
- 10 points: code to test the function